

A Method for Passive Identification of Spacecraft	1
Using Doppler-Based Trajectory Matching	2
Abstract	3
This paper presents a passive method for identifying unknown spacecraft using	4
the Doppler signature of a single satellite pass, requiring neither training data	5
nor the precise knowledge of the transmitter carrier frequency. Doppler shifts	6
are extracted from SDR IQ samples via a phase-locked loop and averaged into	7
a one second time series. Candidates from an open Two-Line Element catalog	8
are processed by a seven-stage filtering algorithm before predicted Doppler tra-	9
jectories are generated using a computationally efficient combination of SGP4	10
propagation and F&G (Lagrange coefficient) functions. Observed and predicted	11
series are independently mean-centered to cancel unknown carrier offsets and	12
SDR oscillator instability, and candidates are ranked by root-mean-square error.	13
Using consumer-grade SDR receivers, the method achieved 28 successful single	14
pass identifications across the VHF, UHF, L-, and S-bands, with the correct	15
satellite ranking first in every case and an RMSE typically several times lower	16
than the nearest competing candidate. The implementation is released as open-	17
source software. These results show that accurate, low-cost passive spacecraft	18
identification is achievable without specialized infrastructure or prior calibration.	19
Keywords: Passive satellite identification, Doppler frequency shift, single pass	20
identification, Software-defined radio, Space situational awareness, Non-cooperative	21
identification, Two-Line Element set, SGP4, F&G functions	22

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56 1 Introduction

57 Low earth orbit has grown significantly more congested over the past decade. The
58 Starlink constellation alone already comprises more than 8,000 satellites and is planned
59 to expand to approximately 42,000 spacecraft within the next decade (Yang et al.
60 2026). A direct consequence is denser artificial radio frequency emissions, which make
61 identification of individual sources substantially harder. When telemetry is unavailable
62 or access is restricted, linking a received signal to a specific spacecraft becomes a
63 nontrivial problem.

64 The existing approaches to this problem can be divided into three broad categories.

65 The first category comprises governmental space surveillance infrastructures, such
66 as the Space Surveillance Network (SSN) and NORAD. They operate as distributed
67 networks of specialized ground-based and orbital observation assets (Kelso 1997).
68 Such systems are unfortunately inaccessible to independent and academic users. The
69 growing number of tracked objects also introduces significant computational chal-
70 lenges associated with correlating uncataloged observations (Phillion et al. 2014). It
71 is important to mention that some systems in this category rely on optical observa-
72 tions, including astrometric tracking and photometric light curve analysis. Since these
73 methods require line-of-sight visibility and depend on favorable illumination condi-
74 tions, they are complementary to RF-based techniques and are outside the scope of
75 this work.

76 The second category consists of radio-frequency fingerprinting (RFF) techniques
77 based on physical imperfections of the transmitter hardware. In restricted scenar-
78 ios, particularly for Iridium and GPS constellations, this approach works well, with
79 reported accuracies of 80–100% when convolutional neural networks are used for clas-
80 sification (Oligeri et al. 2020; Solenthaler et al. 2025). However, RFF methods require
81 the collection of substantial training datasets through extensive measurement cam-
82 paigns. Furthermore, as demonstrated by Solenthaler et al. (2025), such methods

are typically tailored to known signal types, require separate training for each new constellation, and cannot generalize to arbitrary spacecraft in an open catalog.

The third category encompasses methods based on Doppler frequency shift measurements. Their physical generality makes them the most natural starting point for the present work. None of the works reviewed below fully solves the identification problem as defined here. The study "Non-Cooperative LEO Satellite Orbit Determination Using Single Station" ([Deng et al. 2024](#)) performs non-cooperative orbit determination from Doppler measurements collected at a single ground station. However, accurate orbit estimation requires approximately 24 hours of accumulated observations. The work of [Spiridonov et al. \(2021\)](#) addresses the identification of an unknown spacecraft using Doppler measurements and Two-Line Element (TLE) catalog data, but relies on observations spanning multiple orbital revolutions. Finally, passive characterization of spacecraft through analysis of Doppler curves was demonstrated by [Richmond and Raichle \(2018\)](#). Their implementation, however, utilized a decommissioned S-band phased-array antenna system that is not easily accessible for researchers and small satellite operators.

The key gaps are therefore: the absence of a method capable of operating using data from a single satellite pass, the dependence of existing approaches on precise knowledge of the carrier frequency, and the limitation of most solutions to a single frequency band.

Accessible passive monitoring tools are needed across several domains. In radio astronomy, satellite transmissions may exceed the power levels of astrophysical sources by many orders of magnitude. Furthermore, certain classes of radio interference can imitate Doppler-like patterns characteristic of transient astrophysical events, thereby generating false detections in automated discovery systems ([Yang et al. 2026](#)). Small satellite operators frequently encounter difficulties identifying their own satellites following deployment from a launch vehicle: existing onboard identification methods

110 exhibit substantial failure rates ([Skinner et al. 2022](#)). Finally, regulatory monitoring of
111 radio-frequency spectrum usage requires the ability to identify spacecraft transmitting
112 outside their allocated frequency bands.

113 The aim of this work is to develop and experimentally validate a method for passive
114 satellite identification based on the Doppler signature of a single pass while imposing
115 minimal requirements on computational resources and hardware infrastructure.

116 The novelty of the proposed approach is defined by three key aspects. First, unlike
117 orbit determination methods ([Deng et al. 2024](#); [Spiridonov et al. 2021](#)), identification
118 is performed using data from a single pass without the need to accumulate observa-
119 tions over multiple orbital revolutions. Second, the method does not require precise
120 knowledge of the carrier frequency. Systematic errors caused by the instability of the
121 receiver’s and transmitter’s reference oscillators, as well as tuning inaccuracy is elimi-
122 nated through an averaging operation applied during signature matching. As a result,
123 the approach remains applicable to receivers such as the HackRF One and similar
124 consumer-grade software-defined radio (SDR) platforms that do not possess ultra-
125 stable frequency references. Third, unlike RFF-based techniques ([Oligeri et al. 2020](#);
126 [Solenthaler et al. 2025](#)), the proposed method is applicable to any spacecraft repre-
127 sented within an open TLE catalog and does not require prior collection of training
128 data. In addition, a computationally efficient scheme is proposed for generating a cat-
129 alog of predicted signatures through a combination of the SGP4 propagation model
130 and F&G functions (Lagrange coefficient functions of the two-body problem).

131 To achieve the stated objective, the following tasks are addressed:

- 132 1. Development of a method for extracting Doppler signatures from IQ (In-phase and
133 Quadrature) data using a phase-locked loop (PLL).
- 134 2. Construction of a model relating Doppler frequency shift to relative velocity while
135 accounting for acceptable approximations.

3. Development of a computationally efficient orbit propagation approach based on a
combination of SGP4 and F&G functions. 136
4. Design of an algorithm for candidate selection and matching of observed and
predicted trajectories. 139
5. Development and verification of a procedure for compensating receiver reference
frequency instability. 141
6. Experimental validation of the proposed method using real-world observations in
the VHF, UHF, L-, and S-bands. 143

The proposed method can be implemented on consumer-grade SDR hardware,
making passive satellite identification broadly accessible for scientific, engineering, and
regulatory use. 144 145 146

2 Theoretical Foundations of Passive Doppler Satellite Identification 147 148

2.1 Doppler Frequency Shift in Satellite Observation Problems 149

The Doppler frequency shift arises from the relative motion of a radiation source and
an observer along the line of sight. For electromagnetic waves, the observed frequency
increases as the source and observer approach one another and decreases as they
recede. In the classical case the observed frequency is given by: 150 151 152 153

$$f_{\text{obs}} = \frac{c + v_0}{c - v_s} f_{\text{sat}} \quad (1)$$

where f_{obs} is the observed frequency, f_{sat} is the transmitter frequency, v_0 is the
observer speed toward the source, v_s is the source speed toward the observer, and c is
the speed of light. Positive velocities correspond to mutual approach. 154 155 156

For near Earth satellites, since the relative velocity is well below the speed of light,
 $v_{\text{rel}} \ll c$, the first-order approximation can be used: 157 158

$$v_{\text{rel}} \approx \left(\frac{f_{\text{obs}}}{f_{\text{sat}}} - 1 \right) c \quad (2)$$

159 The neglected second-order term, originating from a Taylor expansion of Equation 1,
160 takes the form:

$$\frac{v_s(v_s + v_0)}{c^2} \quad (3)$$

161 Even at the upper end of LEO velocities, where $v_{\text{rel}} \approx 1.2 \times 10^4$ m/s, the
162 resulting error is on the order of 10^{-9} . For an S-band signal at 2.25 GHz this is
163 roughly 3 Hz, which is well below the frequency errors typical of consumer-grade SDR
164 receivers (Bednarz et al. 2024).

165 2.2 Frequency Extraction from SDR Signals

166 Estimating the instantaneous frequency of a signal from complex IQ samples is a well-
167 established problem in digital signal processing. For satellite signals it is addressed in
168 software using SDR receivers through a combination of spectral analysis and phase-
169 tracking techniques (Guo et al. 2015; Gao et al. 2016). A significant practical constraint
170 is the reference oscillator instability of low cost SDR platforms, which can introduce
171 systematic offsets into measured frequency values (Bednarz et al. 2024). The specific
172 methods used in this work to extract Doppler signatures from IQ data are described
173 in Section 3.3.

174 2.3 Orbit Propagation Models

175 The standard format for orbital data on objects tracked by space surveillance networks
176 is the Two-Line Element set (TLE), distributed by organizations such as CelesTrak.
177 Orbit Mean-Element Messages (OMMs) defined by the Consultative Committee for
178 Space Data Systems (CCSDS) provide a modern, machine-readable representation of
179 the same mean orbital elements and associated metadata as TLEs, while avoiding the

ambiguities and formatting limitations of the TLE format ([for Space Data Systems](#) 180
[2009](#)). Orbit propagation from both TLE and OMM data is performed using the SGP4 181
analytical model, which accounts for the principal perturbations acting on a satellite 182
and is the standard in near Earth space object tracking ([Vallado et al. 2006](#)). Due to 183
the simplicity and availability of the TLE element sets during the development stages 184
of the project, the developed program uses TLEs in favor of OMMs. Alongside SGP4, 185
propagation of the orbital state vector may be performed via Lagrange coefficients (the 186
F&G functions) which return the satellite position and velocity at an arbitrary epoch 187
analytically within the two-body framework ([Sharifi and Seif 2011](#); [Seif 2017](#)). The use 188
of these tools to generate predicted Doppler signatures is described in Section 3.4. 189

3 Method for Passive Satellite Identification Based 190 on Doppler Matching 191

3.1 General System Architecture 192

The system operates in two modes: real-time and post-processing. In real-time mode, 193
data is received directly from the SDR receiver via SDR++, a cross platform software- 194
defined radio application ([Rouma 2021](#)). In post-processing mode, the input is a 195
frequency time series file produced during a prior observation, with the option to over- 196
ride timestamps and observer coordinates. Since the two modes differ only in how input 197
data is supplied, the following describes the common processing pipeline, illustrated 198
in Figure 1. 199

In the first stage, complex IQ samples are sent from the IQ Exporter module of 200
SDR++ to the application over TCP and accumulated in an internal buffer. If the 201
signal contains a clear carrier, an optional initial frequency estimate is computed via 202
the Fast Fourier Transform (FFT) and used to initialise the numerically controlled 203
oscillator (NCO). If this FFT step is skipped, the NCO is initialised with zero offset 204

relative to the center frequency of the VFO. The signal is then tracked by a phase-locked loop (PLL). The PLL output samples are averaged over one second intervals to form a frequency time series.

At the same time, TLE sets are retrieved from a local database that is updated in advance to avoid exceeding external API rate limits. Based on the observed frequency at the initial epoch t_0 and specified spatiotemporal constraints, multi-stage candidate prefiltering reduces the object set to a manageable size. Such filtering schemes are widely used in observation-correlation and hypothesis-space reduction problems (Phillion et al. 2014). For each candidate that passes filtering, SGP4 computes the satellite position and velocity at t_0 , after which F&G functions propagate the state to all subsequent observation epochs. The observer position in the GCRS frame at any epoch is obtained by rotating the ITRS vector by the angle corresponding to elapsed time. Predicted Doppler shifts are computed from the satellite-observer relative position vector. The observed and predicted series are independently centered by subtracting their respective means, and an RMSE value is computed for each candidate. The final candidate list is ranked in ascending order of RMSE.

3.2 Hardware and Measurement Setup

3.2.1 SDR Receiver Chain

Three software-defined radio receivers were used in the experimental study. The first is the LibreSDR B210 mini, based on the AD9361 transceiver, with a specified reference oscillator stability of 0.5 ppm. Its architecture is close to that of the USRP B200 mini, whose characteristics are studied in detail in the study by Bednarz et al. (2024). The second is the HackRF One (Clifford R10+ revision) fitted with an external temperature-compensated crystal oscillator (TCXO) specified at 0.1 ppm. The third is the RTL-SDR v3 with a specified oscillator stability of 1 ppm. The HackRF and the B210 mini were used interchangeably for VHF, UHF, L- and S-band observations,

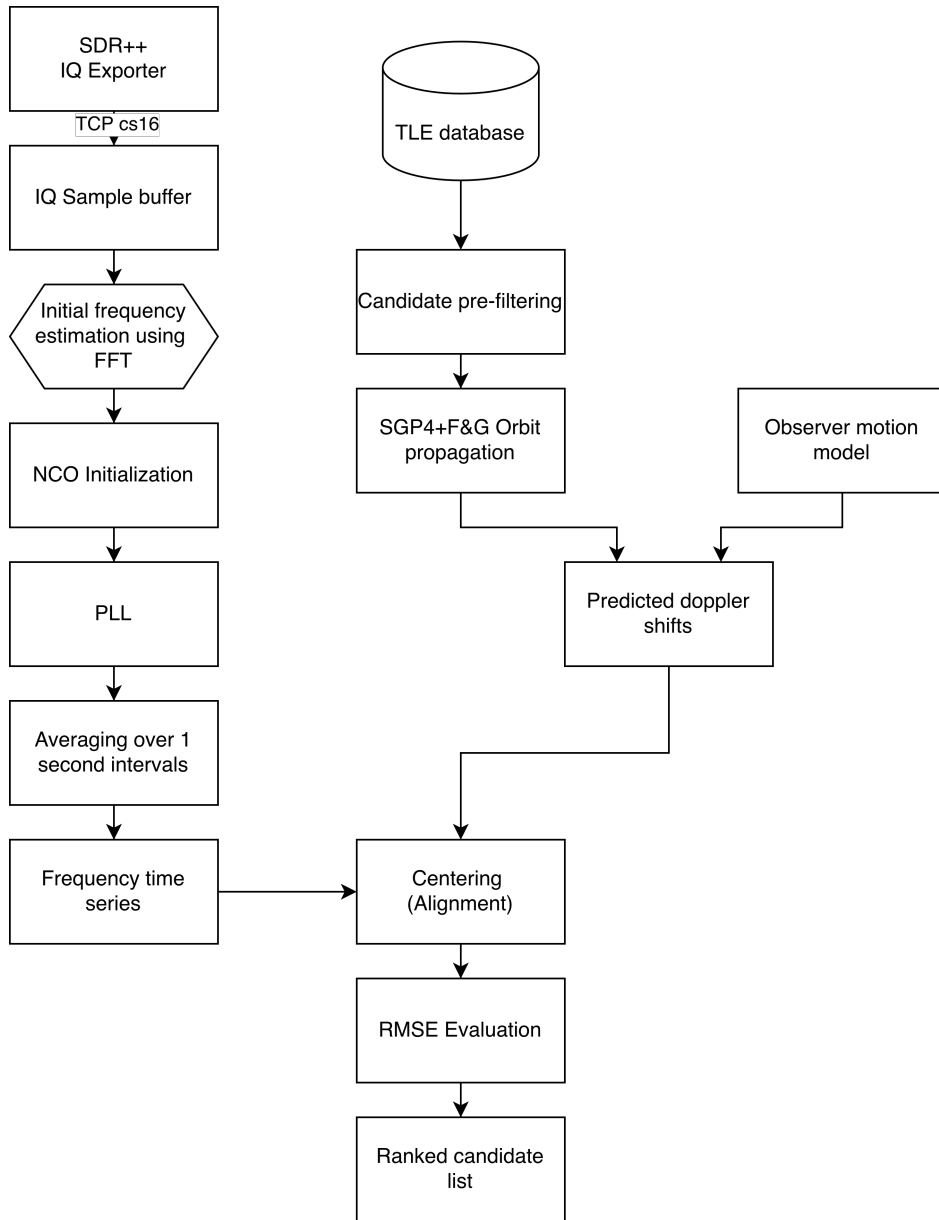


Fig. 1 Generalized processing chain of the passive Doppler satellite identification system.

whereas the RTL-SDR v3 was used in the L band only. No quantitative comparison 231
between receivers was performed within the scope of this work. A GPS-disciplined 232

oscillator (GPSDO) was intentionally excluded from the setup, as one of the objectives of this work is to demonstrate the applicability of the method on widely available consumer hardware.

The antenna front-ends varied by frequency band. In the VHF band a telescopic V-dipole antenna was used; in the UHF band a consumer-grade 7-element Yagi-Uda antenna with a folded dipole was used. Neither configuration included an external low-noise amplifier (LNA).

For the L and S bands an identical reflector, a 0.9m offset parabolic dish (AX-OFFSET D90), was used with different feeds and amplifier chains. In the L band, a left-hand circularly polarised (LHCP) helical feed, designed using an online calculator ([Helical Antenna Calculator 2024](#)), was mounted at the focal point of the reflector. Due to polarisation reversal upon reflection from a parabolic surface ([Kraus and Marhefka 2002](#)), the system provides effective reception of right-hand circularly polarised (RHCP) signals from the source side. The receiving chain comprised a two-stage LNA with a SAW filter optimised for 1690 MHz, followed by a broadband LNA based on the TQP3M9037 transistor. In the S-band, a separate LHCP helical feed designed by the same method was used together with the SeLNA v1 LNA ([SeLNA-v1 2026](#)), a modified ADS-B LNA with replacement bandpass filters adapted for 2200-2300 MHz. The two receiving chains are shown in Figure 2.

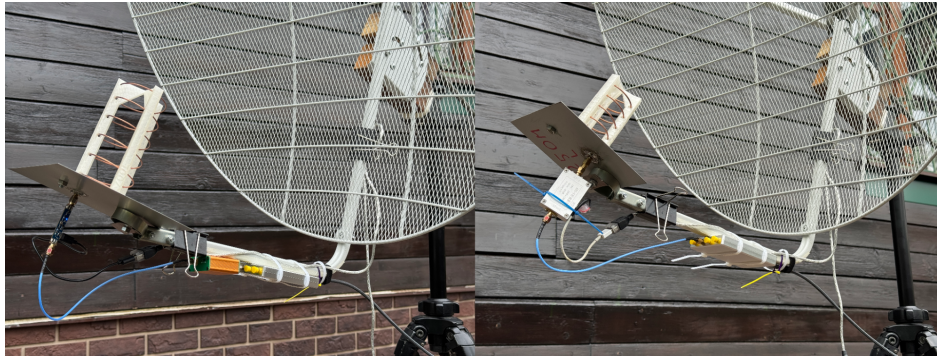


Fig. 2 Photograph of the L-band (left) and S-band (right) receiving chains.

3.2.2 Reference Frequency Stability Assessment 252

A key source of systematic error in Doppler measurements is the instability of the SDR receiver's reference oscillator. As reported by [Bednarz et al. \(2024\)](#), even receivers with specified stability of 0.5–1 ppm exhibit both constant frequency offsets and slow thermal drift. The present experiments confirmed that the frequency readings of both receivers are sensitive to ambient temperature: the best results were obtained under stable thermal conditions, whereas noticeable drift was observed during transient phases such as equipment warm-up.

The presence of a constant frequency offset in both receivers was qualitatively confirmed by briefly observing the downlink signal of the geostationary satellite SBIRS-GEO 1 (SATCAT No. 37481) at 2262.5 MHz. Because the Doppler shift of a geostationary satellite is negligible compared to SDR oscillator instability, any observed frequency offset can be attributed to the systematic error of the receiver reference. This measurement was indicative only and does not constitute a quantitative characterization of oscillator stability.

None of these error sources is eliminated by hardware calibration in the present work. Instead, constant offset is compensated algorithmically through the series-centering operation applied to both the observed and predicted series, as described in Section 3.5.2. This approach removes the dependence of the method on an ultrastable reference oscillator and preserves its operability across a broad class of consumer SDR platforms.

3.3 Doppler Signature Extraction 273

3.3.1 IQ Data Stream and Preprocessing 274

Complex IQ samples are sent to the application from the SDR++ IQ-Exporter module over TCP, with the application acting as client. Samples are transmitted in the

277 `cs16` format (pairs of signed 16-bit integers encoding the I and Q quadrature com-
 278 ponents). The sample rate and center frequency are queried dynamically from the
 279 SDR++ `rigctl` server module via the `rigctl` protocol, eliminating the need for manual
 280 configuration. Incoming samples are accumulated in a fixed-size internal ring buffer
 281 before being passed to subsequent processing stages. No additional filtering or decima-
 282 tion is applied at this stage. The full sample stream is forwarded as-is to the frequency
 283 estimation block.

284 3.3.2 Initial Frequency Estimation via FFT

285 Before the phase-locked loop is started, an optional initial frequency estimate is com-
 286 puted via the Discrete Fourier Transform. This step is intended for signals containing
 287 a distinct carrier and can substantially accelerate PLL lock (Gao et al. 2016). When
 288 enabled, the application accumulates samples for one second and computes the ampli-
 289 tude spectrum via FFT. The spectral bin with the maximum amplitude is taken as
 290 the initial carrier frequency estimate f_0 , which is used to initialise the NCO:

$$f_0 = \arg \max_k |X[k]| \quad (4)$$

291 where $X[k]$ is the complex spectrum of the input sample block. In the event that
 292 this step is skipped, for signals without a distinct carrier where peak finding is not
 293 meaningful for example, the NCO is initialised with zero offset relative to the receiver
 294 center frequency.

295 3.3.3 Signal Tracking via Phase-Locked Loop (PLL)

296 The primary tool for extracting the instantaneous signal frequency is a phase-locked
 297 loop implemented as a C++ module with a Python interface via `pybind11`. Unlike
 298 classical demodulation schemes, this implementation uses the PLL in frequency esti-
 299 mation mode. Here the output is the current estimate of the instantaneous signal

frequency rather than a recovered phase or demodulated data (Guo et al. 2015; Gao et al. 2016). 300 301

Before the loop is initialised, the maximum permissible frequency deviation from the center frequency $f_{\max}$ is computed. This is the range within which the PLL can acquire and track the signal: 302 303 304

$$f_{\max} = \left(\frac{c + v_0}{c - v_s} - 1 \right) \cdot f_{\text{sat}} \cdot M \quad (5)$$

For the worst case (maximum Doppler shift), a satellite at 100 km altitude with $v_s \approx 7850$ m/s and an observer moving toward it at the Earth's equatorial rotation speed $v_0 \approx 465$ m/s, this simplifies to: 305 306 307

$$f_{\max} \approx 2.8 \times 10^{-5} \cdot f_{\text{sat}} \cdot M \quad (6)$$

where $M > 1$ is a user-defined capture-range expansion factor that compensates for frequency tuning inaccuracy and reference oscillator instability in both the receiver and transmitter. 308 309 310

The key parameter of a PLL is the loop bandwidth, which determines the trade-off between signal acquisition speed and noise suppression. A wide bandwidth enables fast lock but requires a higher signal-to-noise ratio, whereas a narrow bandwidth provides accurate tracking of a weak signal but slows initial acquisition (Guo et al. 2015). The implementation provides a set of pre-configured bandwidth values that the operator can switch during an observation. The typical strategy in the L and S bands is to use a 2,000Hz bandwidth for initial lock, then narrow to 300Hz for fine tracking. In the VHF and UHF bands, proportionally smaller values were used, which empirically provided stable tracking. 311 312 313 314 315 316 317 318 319

The PLL output is subject to noise from phase jitter and dynamic tracking error, the dominant source of which is thermal noise in the receiving chain (Kaplan and Hegarty 2006). 320 321 322

323 3.3.4 Formation of the Frequency Time Series

324 To suppress noise in the PLL output stream, a one second averaging procedure is
325 applied. All frequency samples f_i arriving in the interval $[N - 0.5, N + 0.5)$, where N
326 is an integer Unix timestamp, are averaged into a single value:

$$\hat{f}(N) = \frac{1}{|S_N|} \sum_{i \in S_N} f_i, \quad S_N = \{i : |t_i - N| < 0.5\} \quad (7)$$

327 The result is a time series $\{(N_j, \hat{f}(N_j))\}$ representing the Doppler signature of the
328 observed signal sampled at one second intervals. This gets fed into the identification
329 algorithm described in Section 3.5.

330 3.4 Generation of Predicted Doppler Trajectories

331 3.4.1 TLE Acquisition and Processing

332 Two-Line Element sets are obtained from several configurable sources. The first is
333 CelesTrak, which provides the active satellite catalog through an open API. The second
334 is ReTLEctor ([ReTLEctor 2026](#)), a lightweight proxy server that caches CelesTrak
335 data to prevent rate-limit violations during bulk downloads. The third is Space-Track,
336 the official catalog of the US Space Forces Command, from which records with an
337 epoch no older than 14 days and no decay date are requested. The fourth is the
338 Mike McCants archive ([Mike McCants' Satellite Tracking TLE ZIP Files 2026](#)), which
339 contains unofficial TLE data for satellites absent from official catalogs but identified
340 through legitimate passive observations (this source does not redistribute classified
341 data). Finally, a user-supplied plain text file of TLE sets allows arbitrary orbital
342 elements to be included optionally.

343 The retrieved data is stored in a local database. When the same satellite appears in
344 multiple sources, the entry with the most recent epoch is kept. The database is updated
345 manually by the operator before each observing session. Local caching removes the

dependency on external API availability at identification time and avoids latency from
real-time network requests.

3.4.2 Orbit Propagation via SGP4 + F&G Functions

Predicted Doppler trajectory generation is performed in two stages to minimise
computational cost.

In the first stage, for each candidate object that has passed the initial filtering, the
SGP4 model (Vallado et al. 2006) is called once to compute the position and velocity
vector at the initial observation epoch t_0 :

$$\mathbf{r}_{\text{sat}}(t_0), \mathbf{v}_{\text{sat}}(t_0) = \text{SGP4}(\text{TLE}, t_0) \quad (8)$$

If the Doppler pre-filter is enabled (Section 3.5.1), the computed state is used to
calculate one predicted Doppler offset at t_0 and immediately reject candidates that do
not pass, without propagating the full orbit. The result is also used for the visibility
filter. These two steps are the last in the filtering chain.

In the second stage, for objects that pass all filters, the orbit is propagated over
the full observation interval using F&G functions (Sharifi and Seif 2011; Seif 2017).
At each epoch $t_0 + \Delta t$ the new state vector is computed independently from the initial
state $(\mathbf{r}_0, \mathbf{v}_0)$:

$$\mathbf{r}(t_0 + \Delta t) = f \mathbf{r}_0 + g \mathbf{v}_0 \quad (9)$$

$$\mathbf{v}(t_0 + \Delta t) = \dot{f} \mathbf{r}_0 + \dot{g} \mathbf{v}_0 \quad (10)$$

The coefficients $f, g, \dot{f}, \dot{g}$ are computed from the change in eccentric anomaly ΔE ,
which is found from the generalized Kepler equation by Newton-Raphson iteration:

$$n \Delta t = \Delta E - \left(1 - \frac{r_0}{a}\right) \sin(\Delta E) + \frac{\mathbf{r}_0 \cdot \mathbf{v}_0}{\sqrt{\mu a}} (1 - \cos(\Delta E)) \quad (11)$$

365 where a is the semi-major axis, $n = \sqrt{\mu/a^3}$ is the mean motion, and $\mu = 3.986004418 \times$
 366 $10^5 \text{ km}^3/\text{s}^2$ is the Earth's standard gravitational parameter. Iteration terminates when
 367 the convergence criterion $|\delta(\Delta E)| < 10^{-12}$ rad is satisfied, or after a maximum of 10
 368 iterations. Once ΔE is found, the coefficients are evaluated as:

$$f = 1 - \frac{a}{r_0}(1 - \cos \Delta E), \quad g = \Delta t - \sqrt{\frac{a^3}{\mu}}(\Delta E - \sin \Delta E) \quad (12)$$

369

$$\dot{f} = -\frac{\sqrt{\mu a}}{r r_0} \sin \Delta E, \quad \dot{g} = 1 - \frac{a}{r}(1 - \cos \Delta E) \quad (13)$$

370 To reduce computational cost, satellite and observer positions are precomputed and
 371 cached for 1,000 consecutive seconds starting from t_0 , which covers the typical duration
 372 of observable LEO passes. When the Doppler shift at a specific epoch is requested,
 373 the corresponding state vectors are retrieved from the cache without recomputation.

374 3.4.3 Observer Motion Model

375 The precise observer position in the GCRS frame at the initial epoch t_0 is computed
 376 using the Skyfield library, accounting for Earth's precession and nutation. For all
 377 subsequent epochs $t = t_0 + \Delta t$, the observer coordinates are obtained by rotating the
 378 ITRS vector by the angle corresponding to elapsed time:

$$\mathbf{r}_{\text{obs}}(t) = \mathbf{R}_z(\omega_{\oplus} \Delta t) \mathbf{R}_{\text{ITRS} \rightarrow \text{GCRS}}(t_0) \mathbf{r}_{\text{ITRS}} \quad (14)$$

379 where $\omega_{\oplus} = 7.2921150 \times 10^{-5} \text{ rad/s}$ is the Earth's rotation rate. The observer velocity
 380 in the GCRS frame is:

$$\mathbf{v}_{\text{obs}} = \boldsymbol{\omega}_{\oplus} \times \mathbf{r}_{\text{obs}} = \begin{bmatrix} -\omega_{\oplus} y \\ \omega_{\oplus} x \\ 0 \end{bmatrix} \quad (15)$$

This approximation neglects small variations in Earth’s rotation axis orientation 381
over the observation interval, which is well justified for typical LEO pass durations of 382
10-15 minutes. 383

3.4.4 Computation of the Predicted Doppler Shift 384

At each observation epoch the predicted Doppler shift is computed from the projection 385
of the relative velocity vector onto the line of sight. The unit line of sight vector is: 386

$$\hat{\boldsymbol{\rho}} = \frac{\mathbf{r}_{\text{sat}} - \mathbf{r}_{\text{obs}}}{|\mathbf{r}_{\text{sat}} - \mathbf{r}_{\text{obs}}|} \quad (16)$$

The radial approach velocity along the line of sight is: 387

$$\dot{\rho} = (\mathbf{v}_{\text{obs}} - \mathbf{v}_{\text{sat}}) \cdot \hat{\boldsymbol{\rho}} \quad (17)$$

where a positive $\dot{\rho}$ corresponds to mutual approach. Using the first-order approxima- 388
tion from Section 2.1, the predicted observed frequency is: 389

$$f_{\text{pred}} = f_{\text{sat}} \left(1 + \frac{\dot{\rho}}{c} \right) \quad (18)$$

Since the exact transmitter carrier frequency f_{sat} is generally unknown, the abso- 390
lute values of f_{pred} are not used directly in the matching algorithm. Instead, both the 391
observed and predicted series are centered by subtracting their respective means, as 392
described in Section 3.5.2. 393

3.5 Satellite Identification Algorithm 394

3.5.1 Prefiltering of Candidate Satellites 395

The TLE catalog currently contains approximately 15,000 objects, for which full 396
orbit propagation is impractical. To reduce the candidate set to a manageable size, 397

398 sequential multi-stage filtering is applied. The seven filters are ordered by increasing
399 computational cost.

400 The first five filters operate directly on TLE fields without invoking the propaga-
401 tion model. The *debris filter* rejects objects whose name contains the substrings *deb*,
402 *rb*, or *r/b* (case insensitive), eliminating known debris and rocket bodies. The *epoch*
403 *filter* rejects ephemerides older than a configurable threshold (14 days by default), as
404 propagating such old elements with the SGP4 model yields unacceptable prediction
405 accuracy. The *constellation filter* excludes satellites belonging to mega-constellations
406 (by default Starlink and OneWeb), as their signals are largely inaccessible to passive
407 reception in the bands under study. The *GEO and high-orbit filter* rejects objects with
408 mean motion $n < 6$ revolutions per day, a threshold chosen empirically because the
409 expected Doppler shift for such orbits is substantially lower than for LEO objects and
410 carries little to no discriminating information over short observation intervals. The
411 *high eccentricity filter* excludes objects with eccentricity $e > 0.25$, a threshold cho-
412 sen empirically as a compromise between reducing the candidate count and retaining
413 observable objects.

414 For the remaining objects, SGP4 is called once to compute the position and velocity
415 vector at the initial observation epoch t_0 . The result is shared by the two subsequent
416 filters. The *visibility filter* rejects satellites with elevation below the horizon $\theta_{\text{el}} <$
417 θ_{min} (default: 0°) at t_0 , restricting the candidate set to objects that were above the
418 horizon at the start of the observation. Finally, the *Doppler pre-filter* rejects satellites
419 whose predicted Doppler shift at t_0 differs from the observed value by more than a
420 configurable threshold ε :

$$|\Delta f_{\text{pred}}(t_0) - \Delta f_{\text{obs}}(t_0)| > \varepsilon \quad (19)$$

This particular filter is most effective when the center frequency of the signal is not known precisely. In that case, its use is not recommended, as it risks rejecting the true candidate.

3.5.2 Trajectory Normalisation

Direct comparison of the observed and predicted frequency series is not possible for several reasons. First, the exact transmitter carrier frequency f_{sat} is generally unknown, introducing a constant offset between the series. Second, reference oscillator instability in the SDR receiver adds a further systematic offset and slow drift. The constant offset is removed by centering: the mean of each series is subtracted from it:

$$\tilde{f}_{\text{obs},i} = f_{\text{obs},i} - \bar{f}_{\text{obs}} \quad (20)$$

$$\tilde{f}_{\text{pred},i} = f_{\text{pred},i} - \bar{f}_{\text{pred}} \quad (21)$$

The centered series $\tilde{f}_{\text{obs}}$ and $\tilde{f}_{\text{pred}}$ retain only information about the shape of the Doppler curve, which is invariant to constant offsets of any origin. The operation also makes the method robust to imprecise receiver tuning. Formally, assuming:

$$f_{\text{obs}}(t) = f_D(t) + b + \epsilon(t) \quad (22)$$

where $f_D(t)$ is the Doppler component, b is the constant bias (unknown transmitter frequency combined with the SDR local oscillator offset), and $\epsilon(t)$ is noise, subtracting the mean gives:

$$\tilde{f}_{\text{obs}}(t) = f_D(t) - \bar{f}_D + \epsilon(t) - \bar{\epsilon} \quad (23)$$

because the constant term cancels exactly: $b - \bar{b} = 0$.

438 3.5.3 Matching Metric

439 The proximity between the observed and predicted Doppler signatures is measured by
440 the root mean square error (RMSE) of the centered series:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\tilde{f}_{\text{obs},i} - \tilde{f}_{\text{pred},i} \right)^2} \quad (24)$$

441 where N is the number of timestamps common to both series. RMSE was chosen
442 because the Doppler trajectories are time-synchronised and sampled simultaneously,
443 making amplitude domain deviation the primary discriminating feature. Candidates
444 are ranked in ascending order of RMSE, with the lowest value corresponding to the
445 best match. Across the observations collected in this study, the RMSE of the correct
446 object was typically several times lower than that of the second-ranked candidate,
447 with the gap widening as observation duration increased.

448 3.5.4 Computational Complexity of the Algorithm

449 To quantify pipeline performance, 10 random observer locations and 10 random epochs
450 were combined into 100 test cases, each of which was processed by the full pipeline with
451 synthetic data. Mean execution times for the principal stages are shown in Figure 3.

452 Without the pre-filter, orbit propagation is the dominant stage (78.7% of total
453 time), driven by Newton-Raphson iteration of the Kepler equation for each of ~ 177
454 candidate satellites across 1,000 time points. Enabling the Doppler pre-filter reduces
455 the mean number of candidates reaching the F&G propagation stage from 176.6 to
456 4.7, yielding a 32-fold speedup of the propagation stage and a 4.5-fold speedup of the
457 overall pipeline. The number of remaining candidates after each filtering step is shown
458 in Figure 4.

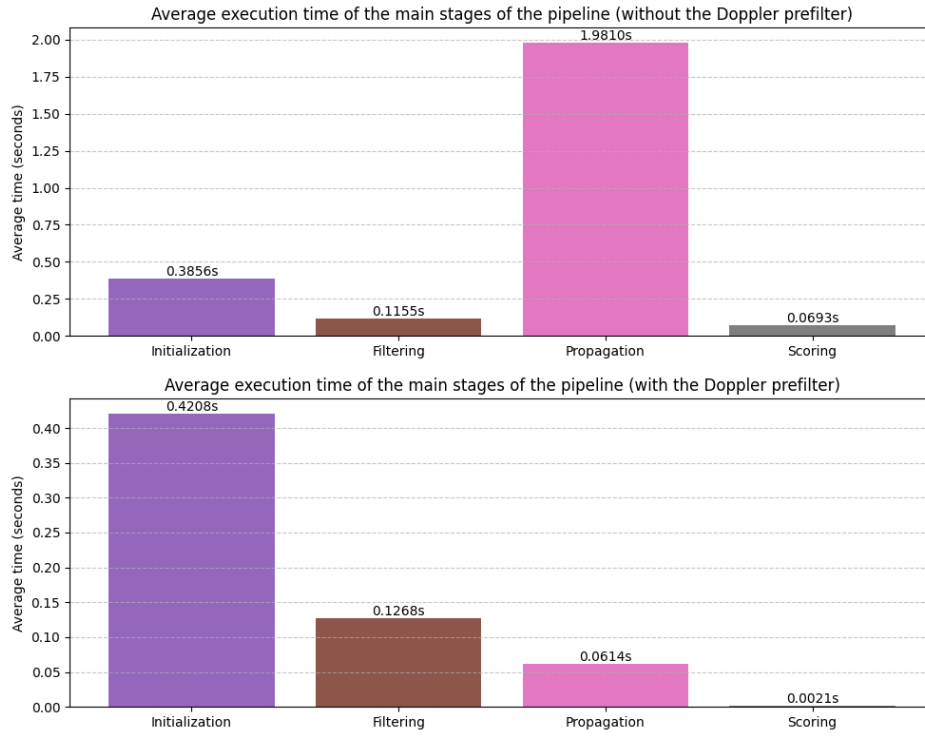


Fig. 3 Execution time breakdown by pipeline stage, with and without the Doppler pre-filter.

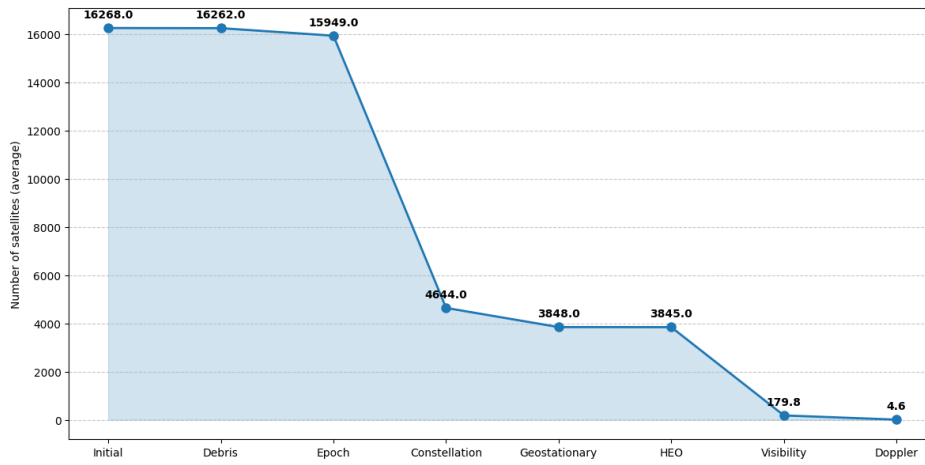


Fig. 4 Candidate count at each stage of the filtering pipeline.

459 4 Experimental Study

460 4.1 Experimental Methodology

461 Observations were conducted near the town of Aprelevka, Naro-Fominsk District,
462 Moscow Region (approximately 55.5°N , 37.1°E), over several weeks in January to
463 February 2026, with a total active observation time of approximately 20 hours. Both
464 receivers described in Section 3.2.1 were used depending on equipment availability at
465 the time of observation. Results did not differ appreciably between receivers.

466 Observations were conducted primarily in the evening, most often around sunset,
467 because of the high density of passes by sun-synchronous satellites during that period.
468 Candidate signals were identified on an opportunistic basis: the operator monitored
469 a live spectrogram in SDR++ looking for unfamiliar signals and checked each one
470 for a pronounced Doppler shift. It appeared as the characteristic leftward drift of
471 spectral lines over time on the FFT plot. Signals showing no visible Doppler shift were
472 disregarded as geostationary or terrestrial.

473 Before each observing session the operator manually updated the local TLE
474 database from CelesTrak, ReTLEctor, and the Mike McCants archive. The last
475 source was added after several early observations yielded no identification because the
476 satellites in question were absent from the official catalogs.

477 The experimental campaign proceeded in two stages. In an initial development and
478 debugging stage, the pipeline was exercised against a broad range of opportunistically
479 observed signals in order to identify edge cases and characterize failure conditions; the
480 three failure modes discussed in Section 4.2 (coherent transponder operation, absence
481 of a satellite from the active TLE catalog, and insufficient signal-to-noise ratio) were
482 identified during this stage and used to refine the observation protocol and candidate
483 filtering. Once the pipeline and protocol had stabilised, a second, formal validation
484 stage was conducted, in which every attempted identification meeting the acquisition

criteria described below was recorded; this stage yielded the 28 observations reported 485
in Section 4.2, all of which were successful. 486

Once a candidate signal was detected, the tripod-mounted antenna was pointed 487
at the source manually. In the UHF, L, and S bands, tracking throughout the pass 488
was maintained by continuously adjusting the antenna direction against the signal 489
level on the FFT display, with a drop in amplitude indicating pointing error. In the 490
VHF band a fixed V-dipole antenna was used without mechanical tracking. The center 491
frequency of the IQ Exporter in SDR++ was set approximately to the observed signal 492
with a downward correction for Doppler shift: since the signal frequency decreases on 493
the descending portion of the pass, the tuning was placed slightly below the observed 494
carrier. The FFT-based initial frequency estimation was enabled where needed. The 495
PLL capture bandwidth was set to 1,000-2,000 Hz for initial lock, then narrowed to 496
300 Hz for fine tracking in the L and S bands, proportionally smaller values were used 497
at lower frequencies. 498

A typical pass lasted approximately 5 minutes, though recorded passes ranged from 499
50 seconds to 10 minutes depending on maximum elevation and observing conditions. 500
The identification algorithm described in Section 3.5 was run live during each pass. 501

Identification correctness was verified in two stages. First, the antenna pointing 502
direction at the time of observation was compared against the predicted azimuth and 503
elevation of the identified satellite in GPredict, where agreement confirmed the plausi- 504
bility of the result. Second, the next pass of the identified satellite was predicted from 505
its TLE and observed independently. Successful tracking with a consistent Doppler 506
curve served as the final confirmation. All 28 observations in the formal validation 507
stage were successful. The failure modes discussed in Section 4.2 were observed during 508
the earlier development stage and are reported for completeness. 509

4.2 Satellite Identification Results

The experimental dataset comprises 28 successful identifications of LEO satellites across the VHF, UHF, L, and S bands: 2 observations in VHF, 1 in UHF, 3 in L, and 22 in S. In every case identification was performed from a single pass, without external receiver frequency calibration and without activating the Doppler pre-filter. Systematic bias was removed solely through the series centering procedure described in Section 3.5.2.

In all observations the correct satellite ranked first by the RMSE metric. Results for a representative sample from each band are given in Table 1.

Table 1 Identification results for a representative sample of satellites.

Satellite	SATCAT No.	Frequency (MHz)	RMSE ₁ (Hz)	RMSE ₂ (Hz)
METEOR-M2 4	59051	137.9	4.39	45.22
IONOSFERA-M 3	64876	150	8.25	174.90
IONOSFERA-M 4	64877	400	5.49	509.64
SARAL	39086	1698.4	50.20	773.46
METEOR-M2 3 (TX1)	57166	1700	40.70	460.24
METEOR-M2 3 (TX2)	57166	1544.5	28.59	1032.61
FENGYUN 3F	57490	2264.65	28.28	1928.28
HINODE (SOLAR-B)	29479	2256.22	24.89	1525.86
METOP-B	38771	2230	75.24	690.29
PROBA-2	36037	2235	268.87	1039.19

Across all bands, the RMSE of the true object is substantially lower than that of the nearest false candidate. In the S band, where the largest number of observations was accumulated ($n = 22$), the mean ratio $\text{RMSE}_1/\text{RMSE}_2$ was 0.073, while the mean ratio $\text{RMSE}_2/\text{RMSE}_3$ was 0.732, indicating a sharply defined minimum of the error function at the correct candidate. A similar pattern holds in the L band ($n = 3$): mean $\text{RMSE}_1/\text{RMSE}_2 = 0.060$ against $\text{RMSE}_2/\text{RMSE}_3 = 0.664$, though due to the small sample size the results are less robust. Averaging across the VHF and UHF bands is not meaningful given the small sample sizes. Individual values for those cases are listed directly in Table 1. The table shows that even when the absolute Doppler shift is

small at lower frequencies, the shape of the trajectory remains sufficient for confident 528
discrimination between spacecraft. 529

The largest number of observations was made in the S band, due to the sheer 530
number of available signals and the high discriminating power of their doppler signa- 531
tures. For FENGYUN 3F, for instance, the gap between $RMSE_1$ and $RMSE_2$ exceeds 532
two orders of magnitude. Results in the L and UHF bands are comparable in qual- 533
ity. In the S band, four satellites of the CENTISPACE constellation were additionally 534
identified on different frequencies, demonstrating that the method generalizes across 535
multiple signals within a single constellation. 536

A notable case is the observation of METOP-B, which was successfully identi- 537
fied at the beginning of its pass during the initial testing and development of the 538
application. Later in the pass, when the spacecraft’s coherent transponder locked 539
onto an uplink from the ground station, the observed downlink acquired a double 540
Doppler shift: the uplink experiences Doppler on the ground-to-space path, and the 541
coherently retransmitted downlink experiences Doppler again on the space-to-ground 542
path ([Jet Propulsion Laboratory 2019](#)). This resulted in an abrupt change in the 543
observed Doppler trajectory, causing the identification algorithm to fail. The case 544
clearly illustrates one of the method’s limitations, discussed below. 545

The effect of TLE age on identification accuracy is illustrated by PROBA-2, for 546
which $RMSE_1$ was 268.87 Hz, which is substantially higher than the values typical 547
of other observations due to the use of outdated ephemerides. Nevertheless, the cor- 548
rect object retained first place in the ranking with $RMSE_2 = 1039.19$ Hz, indicating 549
robustness of the method to moderate ephemeris errors. A corresponding example 550
output is shown in Figure 6. 551

Three characteristic failure modes were identified during the observations. The first 552
is coherent transponder operation, illustrated above by the METOP-B case. The sec- 553
ond is the absence of a satellite from certain TLE catalogs: the most notable examples 554

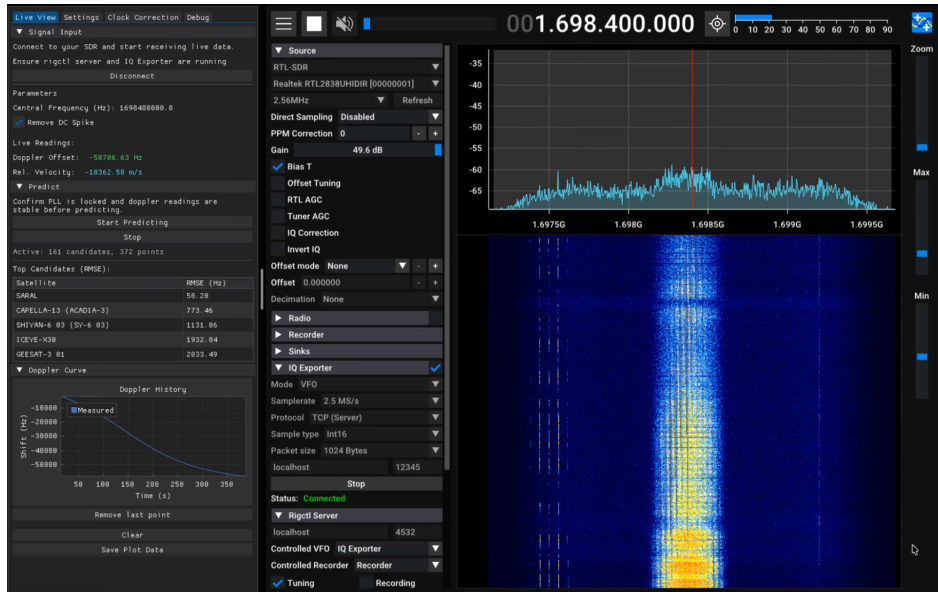


Fig. 5 Identification of SARAL in the L band.

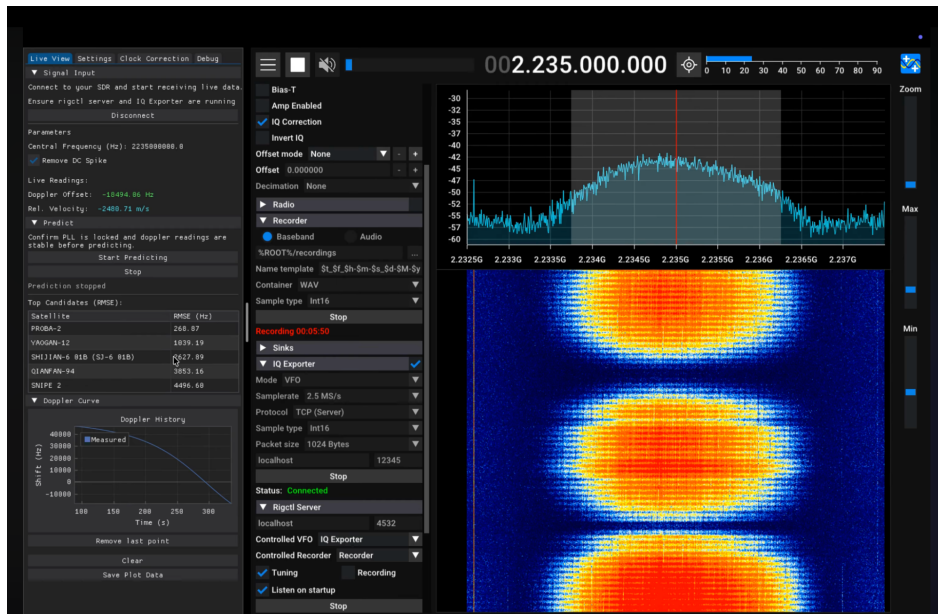


Fig. 6 Identification of PROBA-2 with outdated TLE data.

555 are the NOSS-3 satellites, which do not appear in official open sources but are present
 556 in the Mike McCants archive. In such cases correct identification is impossible without

including that ephemeris source. The third failure mode is insufficient signal-to-noise 557
ratio: under deep signal fades the PLL loses lock, which corrupts the frequency time 558
series. Identification is also degraded for short transmissions, namely brief data-dump 559
sessions or burst-mode transmissions, where the observed Doppler trajectory is too 560
short for confident discrimination between candidates. 561

Taken together, the results confirm that passive satellite identification from a single 562
pass is achievable with consumer grade SDR receivers across a wide frequency range, 563
from VHF through S band, without external frequency calibration and without precise 564
knowledge of the signal carrier frequency. 565

5 Discussion and Conclusion 566

5.1 Main Findings 567

This work presents a method for passive satellite identification from the Doppler sig- 568
nature of a single pass, developed and experimentally validated in full. All objectives 569
stated in the introduction have been addressed. 570

A Doppler signature extraction method was developed based on a phase-locked 571
loop implemented as a high-throughput processing module. The validity of the first- 572
order approximation relating Doppler shift to relative velocity was established for 573
LEO satellite observation scenarios. A computationally efficient approach to gen- 574
erating predicted Doppler trajectories was proposed, combining a single SGP4 call 575
with subsequent orbit propagation via F&G functions. A trajectory matching algo- 576
rithm was developed based on the RMSE metric with mean-subtraction normalisation, 577
which simultaneously compensates for the unknown transmitter carrier frequency, the 578
constant offset of the receiver reference oscillator, and its slow drift. The compensa- 579
tion procedure was thus confirmed to work without external frequency references or 580
GPS-disciplined oscillators. 581

582 The method was experimentally validated on 28 satellites across the VHF, UHF,
583 L, and S bands using consumer-grade SDR receivers. In every case the correct
584 spacecraft ranked first by RMSE with a clear margin over the nearest false candidate.
585 The proposed method has no direct counterpart among the approaches reviewed
586 in the literature. Its implementation on consumer SDR hardware makes it broadly
587 accessible for applications in radio astronomy, post-deployment identification of small
588 satellites, and radio-frequency spectrum monitoring.

589 5.2 Directions for Further Development

590 The most immediate avenue for development is automation of the receiving chain
591 through integration of the antenna with a motorised rotator driven by predicted
592 orbital elements or signal to noise ratio of the unidentified signal in real time. This
593 would replace manual satellite tracking with a fully autonomous observation mode
594 and remove the dependence of signal quality on operator input.

595 A significant functional extension would be support for burst transmissions. The
596 current implementation assumes a continuous signal with a stable carrier suitable for
597 PLL tracking. Adapting the method to short burst transmissions would require an
598 alternative frequency estimation block - for instance, frame by frame FFT followed by
599 aggregation of the individual frequency estimates into a single Doppler curve.

600 A promising direction for improving identification robustness is the inclusion of
601 antenna pointing angles as an additional observable. If the antenna direction corre-
602 sponding to the signal maximum is recorded throughout a pass, the resulting angular
603 trajectory can serve as an independent constraint during candidate matching: satel-
604 lites whose predicted angular trajectory diverges from the observed one can be rejected
605 even when their Doppler RMSE is low. Combining Doppler and angular observables
606 could substantially reduce the probability of false identification in high orbital density
607 environments.

List of abbreviations.	608
<i>ADS-B</i> Automatic Dependent Surveillance-Broadcast	609
<i>API</i> Application Programming Interface	610
<i>CCSDS</i> Consultative Committee for Space Data Systems	611
<i>FFT</i> Fast Fourier Transform	612
<i>GCRS</i> Geocentric Celestial Reference System	613
<i>GEO</i> Geostationary Earth Orbit	614
<i>GPS</i> Global Positioning System	615
<i>GPSDO</i> GPS-Disciplined Oscillator	616
<i>IQ</i> In-phase and Quadrature	617
<i>ITRS</i> International Terrestrial Reference System	618
<i>LEO</i> Low Earth Orbit	619
<i>LHCP</i> Left-Hand Circular Polarisation	620
<i>LNA</i> Low-Noise Amplifier	621
<i>NCO</i> Numerically Controlled Oscillator	622
<i>NORAD</i> North American Aerospace Defense Command	623
<i>OMM</i> Orbit Mean-Element Message	624
<i>PLL</i> Phase-Locked Loop	625
<i>RFF</i> Radio-Frequency Fingerprinting	626
<i>RHCP</i> Right-Hand Circular Polarisation	627
<i>RMSE</i> Root Mean Square Error	628
<i>SATCAT</i> Satellite Catalog Number	629
<i>SAW</i> Surface Acoustic Wave	630
<i>SDR</i> Software-Defined Radio	631
<i>SGP4</i> Simplified General Perturbations Model 4	632
<i>SSN</i> Space Surveillance Network	633
<i>TCP</i> Transmission Control Protocol	634

635 *TCXO* Temperature-Compensated Crystal Oscillator

636 *TLE* Two-Line Element (set)

637 *UHF* Ultra-High Frequency

638 *USRP* Universal Software Radio Peripheral

639 *VHF* Very-High Frequency

640 *VFO* Variable Frequency Oscillator

641 **Declarations.**

642 **Availability of data and materials.**

643 **Competing interests.**

644 **Funding.**

645 **Authors' contributions.**

646 **Acknowledgements.**

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